From Forecasts to Decisions: Benchmarking Probabilistic Demand Forecasts for Retail Replenishment and Liquidation Across Settings and Periods

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ABSTRACT

Public retail forecasting benchmarks rank methods by forecast error, although forecasts determine replenishment quantities. We compare seven probabilistic forecasting methods (three statistical, four gradient-boosting) under one order-up-to policy with optional liquidation, on all eligible series of M5 (physical retail) and VN1 (multi-vendor e-commerce) over three 26-week test windows. Without cost data, methods are compared by the inventory needed to reach the same realised fill rate, with bootstrap confidence intervals.

On M5, LightGBM quantile regression (LGB-Q) is the most accurate method and needs 3–11% less inventory than point forecasts with safety stock, conformal quantiles, TSB-NB and exponential smoothing at a fill rate of 0.94, in every window; the advantage holds at lead times of one to four weeks and under four hyper-parameter settings. On VN1 the best method varies by window: TSB-NB needs 13–16% less inventory on stable-demand series but under-serves growing ones, and neither ADI–CV^2 classes nor an unsupervised segmentation anticipates which series will grow. A post hoc equal-weight average of LGB-Q and TSB-NB quantiles needs 6–11% less inventory than LGB-Q on VN1 at a fill rate of 0.92, stays within 1.5% on M5 and has the smallest worst-case regret (0.9% vs 9.4%), pending independent validation. A univariate N-HiTS baseline leads in no run. Quantile-based liquidation costs at most 0.02 percentage points of fill rate but shows no economic benefit within 26 weeks under the assumed costs.

CCS CONCEPTS • Applied computing → Forecasting; Supply chain management • Computing methodologies → Machine learning; Cluster analysis

Additional Keywords and Phrases: probabilistic forecasting; inventory management; intermittent demand; quantile regression; demand segmentation; liquidation; M5; VN1

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1 INTRODUCTION

Insufficient stock leads to lost sales, while excess stock increases holding requirements and may eventually require liquidation. Inventory decisions therefore depend on the distribution of future demand rather than its point estimate alone. Quantile forecasts provide inputs for replenishment decisions at specified service targets [3].

Public forecasting benchmarks primarily assess prediction accuracy. The M5 competition evaluated weighted RMSSE and weighted scaled pinball loss [1], and its organisers did not define a particular downstream decision problem [3]. Evaluations of the VN1 e-commerce dataset have likewise addressed forecast accuracy and computational cost [4]. These metrics do not necessarily rank methods by inventory performance [5, 6, 23]. Using M5, Theodorou et al. [7] show that accuracy rankings and inventory-cost rankings can diverge, particularly when lost sales are expensive and demand is intermittent. They also call for evaluation on online retail data.

Several aspects remain insufficiently tested. M5 inventory studies often evaluate selected subsets, including 8,000 high-volume series [8], volatile food products [9], or series with at least 2.5 years of history [7]. Such filters exclude some low-volume and intermittent series. Existing studies also commonly use a single dataset and test period [5, 7–9, 23], leaving variation across retail settings and periods less well documented. Finally, public datasets lack observed inventory positions and unit costs. Cost-based assessments must therefore assume these quantities, and conclusions can depend on the assumed cost structure [7].

We evaluate probabilistic forecasting methods by simulating their replenishment and liquidation decisions under a common policy; no new forecasting model is proposed. Specifically, this study (1) compares seven forecasting methods on all eligible series in two retail datasets using a CPU-based experimental pipeline; (2) assesses replenishment at equal realised fill rates with bootstrap confidence intervals, and reports break-even salvage ratios for liquidation; (3) repeats the experiments across three 26-week windows per dataset and checks hyper-parameters, lead times and a neural baseline; (4) examines how observed demand changes relate to rankings on VN1 and tests pre-test demand groupings for method selection; and (5) evaluates, post hoc, an equal-weight combination of the LGB-Q and TSB-NB quantiles. The last analysis is exploratory rather than an independently validated selection rule.

2 RELATED WORK

Retail forecasting benchmarks. LightGBM was used by all top 50 teams in the M5 accuracy track, and the winning solution combined LightGBM models fitted with a Tweedie loss [2]. The leading uncertainty-track solution trained distinct LightGBM models for the forecast quantiles [3]. These approaches inform our direct-quantile and point-forecast baselines. Zanotti [4] analysed the computational and forecast-accuracy effects of combining global models on M5 and VN1, without simulating inventory decisions.

Intermittent demand. Teunter, Syntetos, and Babai (TSB) update the probability of non-zero demand each period to respond to potential obsolescence [10]. The average inter-demand interval (ADI) and squared coefficient of variation of demand size (CV^2) are standard descriptors of intermittency [11]; ADI = 4/3 and $CV^2 = 0.5$ define the class boundaries used for M5 [1]. Damato et al. [12] compared local and global probabilistic methods on intermittent series and found no single established global architecture for this setting.

Forecasts and inventory decisions. Goltsos et al. [6] report that forecast evaluation often omits the subsequent replenishment decision. Kourentzes et al. [13] optimise exponential smoothing against an inventory-oriented criterion, while van der Haar et al. [14] estimate order quantities directly. Petropoulos et al. [24] simulated an order-up-to policy using rolling forecast origins on M3 series and found benefits from forecast combinations. On M5, Spiliotis et al. [23] evaluated a 28-day daily-review simulation and found that simple empirical quantiles could outperform machine-learning approaches in inventory terms. Wang et al. [5] investigated probabilistic combinations under a single-period newsvendor formulation. Zabraoui et al. [9] simulated reinforcement-learning inventory policies for selected products, and Mohammed et al. [8] assessed safety stock with an illustrative cost calculation on high-volume series.

Theodorou et al. [7] provide the closest comparison. Their study simulates one year of lost-sales inventory control for 7,783 daily M5 product–state series, 13 forecasting methods, and 24 combinations of service target, review period, and lead time. Forecast quantiles are generated from point predictions and empirical residual quantiles. Their trade-off curves are broadly aligned with accuracy, whereas costs that penalise lost sales more heavily can favour less accurate models.

Our comparison extends the decision-oriented evaluation to two retail settings, all eligible demand classes, a liquidation decision, and three test windows. We also include directly estimated quantiles and compare the inventory required to achieve the same realised fill rate. This differs from comparing methods at a common nominal service target, which can produce different realised fill rates [7]. Multi-period inventory simulation and decision-based evaluation themselves have been studied previously [5, 7, 9, 14].

3 METHOD

3.1 Data

The M5 dataset [1] records daily unit sales for 3,049 products at 10 Walmart stores, producing 30,490 product–store series. We aggregate daily observations into 277 complete weeks. VN1 [21] comprises 15,053 client–warehouse–product series over 196 weeks and represents multi-vendor e-commerce; its name does not indicate a Vietnamese dataset. A series is eligible for a given test window if its first observation precedes the window by at least 13 weeks and it has at least one sale before the window.

Table 1. Dataset properties. Demand classes and zero-week shares are calculated from histories preceding the main test window.

Property	M5	VN1
Retail setting	One retailer; 10 physical stores	Multi-vendor e-commerce; 46 vendors; 328 warehouses
Total series	30,490	15,053
Eligible series (main / second / third)	30,381 / 29,917 / 28,824	13,844 / 11,442 / 9,383
Number of weeks	277	196
Zero-demand weeks since series start	24.4%	49.1%
Smooth / erratic / intermittent / lumpy	53.0 / 5.8 / 33.9 / 7.3%	18.8 / 13.7 / 47.2 / 20.3%
Additional variables	Prices, events, SNAP days, product hierarchy	Prices in weeks with sales only

Both datasets are processed by the same code. The definition of series start differs between the two source datasets, although the resulting start dates coincide for 99.5% of M5 series. The demand distributions also differ: a classifier using nine demand characteristics distinguishes the datasets with an AUC of 0.91 (Section 5.7).

3.2 Forecast targets and methods

At each weekly forecast origin, the methods predict quantiles $q \in \{0.50, 0.80, 0.90, 0.95, 0.99\}$ of cumulative demand over h weeks. We use h = 3 for replenishment and h = 13 for liquidation. Each cumulative-demand quantile is estimated directly rather than obtained by adding weekly quantiles, since quantiles are not additive.

EMP: empirical quantiles of rolling h-week cumulative demand from the preceding 104 weeks.

ETS: simple exponential smoothing, with a normal approximation to cumulative demand over h weeks [18].

TSB-NB: the TSB approach [10] with a negative-binomial predictive distribution [25] matched to the estimated mean and variance of cumulative demand.

LGB-T: a global LightGBM point forecast using a Tweedie loss [15], combined with a normal safety-stock adjustment estimated from series-level validation residuals.

LGB-C: the same LightGBM point model combined with split-conformal quantiles [17], using residuals pooled by demand class.

HGB-Q: histogram gradient boosting [19] trained with quantile loss on a 300,000-row sample, providing a separate implementation for comparison with LightGBM quantile regression.

LGB-Q: LightGBM quantile regression [16] with one global model per quantile; the main forecasting model in this study.

Section 5.5 additionally examines C2, the arithmetic mean of corresponding LGB-Q and TSB-NB quantiles. The four global gradient-boosting models use common features: lagged sales, rolling means and standard deviations, zero-demand share, weeks since the last sale, calendar variables, price, and series scale. Targets are scaled by the series level. The M5 models also use events, SNAP days, and product hierarchy. Hyperparameters are held fixed and not tuned; Section 5.8 reports their sensitivity.

3.3 Inventory decision policy

Inventory is simulated separately for each series with weekly review and lost sales. In each week: (1) orders placed L weeks earlier arrive; (2) inventory position IP is computed as on-hand plus on-order units; (3) units above the liquidation threshold T are liquidated, where applicable; (4) the policy orders $\max(S - IP, 0)$, with S equal to the τ-quantile of demand over L + 1 weeks, unless liquidation has occurred; and (5) demand is realised, with unsatisfied units counted as lost sales. The default lead time is L = 2 weeks.

Four liquidation policies are compared using the same replenishment forecasts: no liquidation; quantile liquidation, where T is the method's 0.95-quantile of 13-week cumulative demand; fixed liquidation, where T equals 26 times the mean weekly sales over the preceding 26 weeks; and dead13, where a series with no sale for 13 weeks is completely liquidated and replenishment is suspended until the series records another sale.

3.4 Evaluation metrics

Forecast accuracy. The conventional per-series scaled quantile loss divides pinball loss by the in-sample naïve forecast error. Its cross-series mean is unstable on VN1 (Section 5.1). We therefore report volume-weighted quantile loss (wQL), calculated as total pinball loss divided by total demand, as well as the median per-series scaled loss. Mean per-series method ranks are compared using the Friedman test and the Nemenyi critical difference [20].

Inventory performance. Inventory and fill rates are measured over 22 weeks after a four-week warm-up within each test window. Fill rate equals units sold divided by units demanded; inventory is expressed in weeks of demand. Because the methods reach different fill rates at the same target quantile τ, we evaluate τ at each of the five quantile levels and interpolate the inventory required to reach a specified realised fill rate along the inventory–fill-rate curves [7, 25]. Frontier rank is the mean rank of required inventory across fill-rate targets from 0.90 to 0.98. Confidence intervals are computed using 200 bootstrap resamples of series; they quantify cross-series sampling variation, not uncertainty over future time periods.

Liquidation. For each policy, we calculate the break-even salvage ratio s*: the minimum resale or salvage price, as a fraction of unit purchase cost, for which liquidation is at least as profitable as retaining inventory under the assumed costs. The sensitivity analysis uses annual holding-cost rates of 10–40% and gross margins of 30–100%. The datasets contain no observed inventory-cost or salvage-price records.

4 EXPERIMENTAL SETUP

Each dataset is evaluated on its final 26 weeks, divided into two 13-week blocks. Models are refitted at the start of each block; forecasts and decisions are generated weekly. The evaluation is repeated after excluding the final 26 weeks and again after excluding the final 52 weeks, yielding three distinct 26-week windows per dataset (main, second, and third). Model fitting is repeated for each window. All experiments ran on a laptop CPU (Intel Core i5-12450H, 16 GB RAM). Code and results: https://github.com/anhduyBA/forecast-to-decision-benchmark.

5 RESULTS

5.1 Forecast accuracy

Table 2. Forecast accuracy for h = 3 across all eligible series. Windows are main, second, and third. Lower wQL, median loss, and mean rank indicate better performance.

Method	Window	M5 wQL	M5 median	M5 rank	VN1 wQL	VN1 median	VN1 rank
EMP	main	0.115	0.174	4.85	0.172	0.202	4.72
EMP	second	0.120	0.176	4.88	0.146	0.207	4.91
EMP	third	0.126	0.176	4.95	0.197	0.239	4.90
TSB-NB	main	0.088	0.155	4.22	0.129	0.152	3.25
TSB-NB	second	0.091	0.150	3.99	0.088	0.150	3.30
TSB-NB	third	0.098	0.150	4.12	0.137	0.186	3.31
ETS	main	0.084	0.153	4.12	0.129	0.174	4.50
ETS	second	0.089	0.152	4.09	0.092	0.173	4.54
ETS	third	0.095	0.150	4.12	0.136	0.198	4.28
LGB-T	main	0.083	0.152	3.87	0.121	0.188	4.68
LGB-T	second	0.088	0.151	3.90	0.086	0.166	4.45
LGB-T	third	0.094	0.148	3.89	0.126	0.213	4.51
LGB-C	main	0.086	0.159	4.50	0.120	0.184	4.43
LGB-C	second	0.091	0.160	4.60	0.091	0.173	4.48
LGB-C	third	0.096	0.155	4.48	0.125	0.231	4.65
HGB-Q	main	0.077	0.146	3.35	0.104	0.153	3.23
HGB-Q	second	0.081	0.145	3.38	0.079	0.140	3.19
HGB-Q	third	0.087	0.142	3.29	0.104	0.184	3.20
LGB-Q	main	0.076	0.145	3.10	0.103	0.152	3.18
LGB-Q	second	0.080	0.144	3.17	0.078	0.140	3.13
LGB-Q	third	0.086	0.142	3.15	0.102	0.184	3.14

Nemenyi critical difference (α = 0.05), main / second / third: M5 = 0.052 / 0.052 / 0.053; VN1 = 0.077 / 0.084 / 0.093. At h = 13, mean per-series ranks are: M5—LGB-Q 3.45 / 3.43 / 3.50, HGB-Q 3.38 / 3.45 / 3.50, TSB-NB 4.16 / 4.04 / 4.05; VN1—LGB-Q 4.02 / 3.48 / 3.85, HGB-Q 3.85 / 3.46 / 3.57, TSB-NB 2.92 / 2.99 / 3.37.

At h = 3, LGB-Q produces the lowest volume-weighted quantile loss on M5 and VN1 in all three windows; HGB-Q differs by no more than 0.002 (Table 2). Within every demand class, LGB-Q also has the lowest wQL in each dataset and window. On M5, LGB-Q has the lowest mean per-series rank at h = 3 in every window, with differences from HGB-Q exceeding the Nemenyi critical difference.

The per-series results on VN1 are less decisive. LGB-Q, HGB-Q, and TSB-NB differ by at most 0.17 mean ranks at h = 3, and the LGB-Q versus HGB-Q difference is below the critical difference. At h = 13, TSB-NB has the lowest mean per-series rank in each window. Consequently, the global models' advantage in aggregate weighted loss does not imply a uniform advantage across VN1 series. The mean scaled loss for LGB-Q on VN1 increases from 0.336 to 0.539 and 2.008 across the three windows, driven by a small number of series with very small naïve-error denominators; the corresponding wQL values are 0.103, 0.078, and 0.102.

5.2 Inventory at equal realised fill rate

M5. LGB-Q has the lowest frontier rank in every bootstrap resample across all three windows. At a realised fill rate of 0.94, it requires 3.3–10.6% less inventory than LGB-T, LGB-C, TSB-NB, and ETS, with the reported confidence intervals excluding zero (Table 3). HGB-Q is within 0.2–1.4% of LGB-Q.

VN1. Rankings change between windows. LGB-Q has the lowest frontier rank in the main window, but its inventory advantage over LGB-T, LGB-C, and TSB-NB at a fill rate of 0.94 is not statistically resolved by the reported intervals. TSB-NB leads in the second window, with TSB-NB, LGB-T, and LGB-C requiring 4.1–8.8% less inventory than LGB-Q. In the third window, LGB-Q leads, while TSB-NB, ETS, and LGB-T fail to reach a fill rate of 0.94 at the evaluated quantile levels. Thus, the method with the lowest weighted forecast loss does not consistently require the least inventory on VN1.

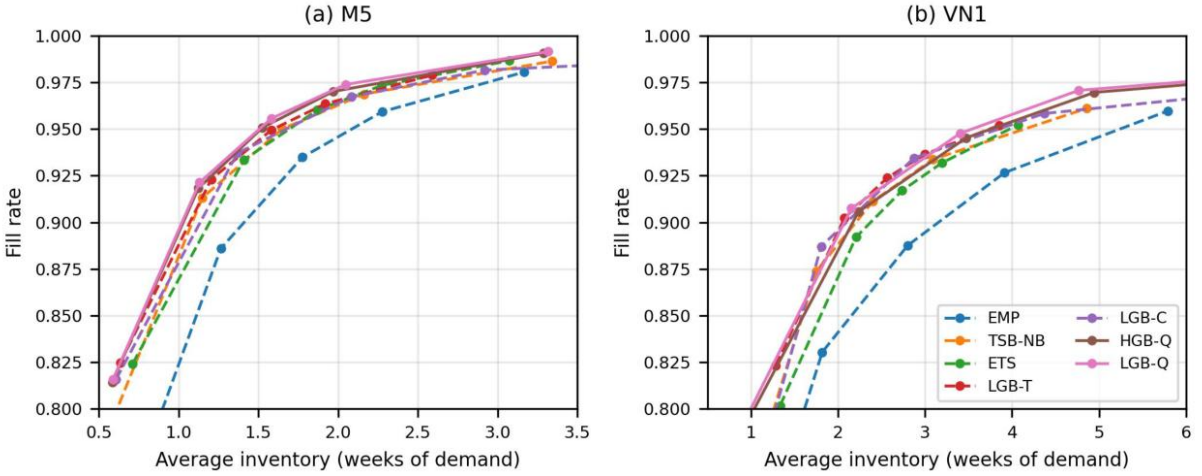


Figure 1. Average inventory (weeks of demand) versus realised fill rate for the main test window: (a) M5 and (b) VN1. The target τ varies from 0.50 to 0.99. Curves toward the upper left indicate less inventory for a given fill rate. Solid lines denote direct quantile models.

Table 3. Inventory required at fill rate 0.94 (weeks of demand) and relative differences versus LGB-Q. Values in brackets are 95% bootstrap confidence intervals. Δ = I_method / I_LGB-Q − 1; positive values favour LGB-Q. "—" indicates that fill rate 0.94 was not reached.

Set	Window	LGB-Q Inv.	Frontier rank	P(best)	HGB-Q Δ %	LGB-T Δ %	LGB-C Δ %	TSB-NB Δ %	ETS Δ %
M5	main	1.38 [1.34, 1.41]	1.00	1.00	+1.2 [+0.9,+1.6]	+5.1 [+4.4,+5.9]	+4.5 [+3.2,+5.8]	+10.0 [+9.1,+11.0]	+10.6 [+9.8,+11.5]
M5	second	1.49 [1.45, 1.53]	1.20	1.00	+1.4 [+1.2,+1.7]	+6.8 [+6.2,+7.6]	+8.2 [+6.8,+9.3]	+7.5 [+6.7,+8.3]	+8.0 [+7.1,+8.9]
M5	third	1.74 [1.69, 1.78]	1.00	1.00	+0.2 [-0.2,+0.5]	+3.3 [+2.7,+4.0]	+7.5 [+6.3,+8.4]	+4.9 [+4.2,+5.7]	+5.9 [+5.0,+6.6]
VN1	main	3.17 [2.78, 3.57]	1.40	0.93	+4.5 [+2.9,+6.7]	+0.9 [-7.5,+9.9]	+1.7 [-6.3,+8.7]	+10.5 [-0.5,+19.8]	+12.0 [+3.8,+19.7]
VN1	second	1.79 [1.52, 2.05]	3.80	0.00	-1.2 [-2.5,+0.2]	-7.7 [-11.8,-3.8]	-4.1 [-7.3,-0.9]	-8.8 [-12.8,-4.9]	-0.4 [-5.7,+4.2]
VN1	third	3.31 [2.72, 3.85]	1.25	0.95	+1.3 [-0.7,+3.5]	—	+11.5 [+1.6,+116.5]	—	—

Across demand classes, LGB-Q has the lowest inventory requirements for the smooth, intermittent, and lumpy M5 classes in all windows. On VN1, only the lumpy class is led by LGB-Q throughout; TSB-NB leads the smooth class in two windows, and LGB-C leads it in the third.

5.3 Period variation on VN1

Table 4. Ex-post demand groups, defined using realised demand in the test period relative to the preceding 26 weeks: stable = 0.67–1.50× and growing > 1.50×. Inventory differences at fill rate 0.92 are Δ = I_TSB-NB / I_LGB-Q − 1. Positive Δ favours LGB-Q; "not reached" indicates TSB-NB cannot attain 0.92.

Dataset	Window	Growing share of test demand	Stable Δ %	Growing Δ %	All series Δ %
M5	main	20.7%	+8.5	+12.5	+10.4
M5	second	28.0%	+7.7	+7.7	+5.5
M5	third	19.4%	+3.9	+5.7	+6.1
VN1	main	31.5%	−14.4	TSB-NB not reached	+5.0
VN1	second	22.8%	−13.3	−10.5	−11.3
VN1	third	40.4%	−15.6	TSB-NB not reached	+25.1

For each window we classify series by the ratio of realised test-period demand to demand in the preceding 26 weeks. Stable-demand series have ratios from 0.67 to 1.50; growing-demand series exceed 1.50. Because the classification uses test-period outcomes, it is descriptive and cannot be used directly for prospective model selection.

On VN1 stable-demand series, TSB-NB requires 13.3–15.6% less inventory than LGB-Q across the three windows (Table 4). For growing-demand series, TSB-NB does not achieve a fill rate of 0.92 in the main and third windows, while LGB-Q does. Growing series account for 31.5% and 40.4% of test demand in those two windows, compared with 22.8% in the second window, when TSB-NB leads overall. This association is consistent with demand-growth composition contributing to the aggregate ranking, but it does not establish a causal mechanism. On M5, LGB-Q requires less inventory in both reported groups in all three windows.

5.4 Prospective method selection using demand groups

Table 5. Method-selection rules fitted in one window and applied to the next. Entries show inventory change (%) at fill rate 0.94 relative to always using LGB-Q; negative values indicate lower inventory. Brackets contain 95% bootstrap confidence intervals.

Selection rule	M5 third → second	M5 second → main	VN1 third → second	VN1 second → main
Always TSB-NB	+7.5 [+6.7,+8.6]	+10.0 [+8.9,+10.9]	−8.8 [−12.6,−5.1]	+10.5 [+1.9,+20.4]
Best single method in previous window	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	+10.5 [+1.9,+20.4]
By ADI–CV² class	+0.1 [−0.0,+0.2]	+0.0 [+0.0,+0.0]	−5.6 [−8.1,−3.9]	+5.1 [−0.3,+10.2]
By segment	+0.1 [−0.1,+0.3]	+0.0 [+0.0,+0.0]	−0.1 [−0.7,+0.5]	+10.5 [+1.9,+20.4]
By class × segment	+0.1 [−0.0,+0.3]	+0.0 [+0.0,+0.0]	−5.6 [−8.1,−3.8]	+4.8 [−0.1,+12.0]
Oracle by segment (not deployable)	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	−8.9 [−12.8,−5.2]	+0.0 [+0.0,+0.0]

Method selection by group would be useful if demand changes could be anticipated from historical characteristics. We test ADI–CV² classes and an unsupervised segmentation based on eight pre-test features: demand level, order size, ADI, CV², time since last sale, trend, recent relative level, and peak size. Clustering is fitted on the earliest window and applied without refitting to later windows. A three-component Gaussian mixture, labelled regular, fading, and sparse, is chosen from K-means and mixture candidates using predefined criteria. Its adjusted Rand index across refits is 0.94, although group separation is weak (silhouette = 0.21).

Neither grouping yields consistent improvements (Table 5). On M5, selecting each group's previously best method changes inventory by at most 0.3% relative to LGB-Q. On VN1, class-based selection reduces inventory by 5.6% in one transfer but increases it by 5.1% in the next. Segment-based selection changes inventory by −0.1% and +10.5%, respectively. An ex-post oracle selecting the best method in each segment does not outperform the best single method for that same window; within-window leading methods are shared across the segments. The proportion of growing series remains broadly distributed across groups: 18–31% for the three segments and 14–39% for the four ADI–CV² classes in VN1.

5.5 Inventory performance of averaged quantiles

LGB-Q and TSB-NB show different inventory outcomes across VN1 demand groups. We therefore examine C2, an equal-weight mean of corresponding LGB-Q and TSB-NB quantiles, without predicting demand growth. This model pair was chosen after inspecting the component-method results; the reported C2 evaluation is thus post hoc and not an independent out-of-sample validation.

At a realised fill rate of 0.92, C2 requires 5.9%, 10.5%, and 7.1% less inventory than LGB-Q on VN1 in the main, second, and third windows, respectively. All reported bootstrap intervals exclude zero; in the second window, C2 matches TSB-NB. On M5, C2 remains within 1.5% of LGB-Q. Its volume-weighted quantile loss is slightly above that of LGB-Q. Adding HGB-Q to the combination produces similar reported outcomes. C2 also requires less inventory than LGB-Q on VN1 at a fill rate of 0.92 with lead times of one and four weeks in every window (Section 5.8).

5.6 Liquidation outcomes

Table 6. Liquidation policies using LGB-Q forecasts. Results are listed separately for the main, second, and third test windows. Changes are relative to no liquidation. s is the break-even salvage ratio (upper-bound range across holding costs of 10–40% per year, gross margins of 30–100%, and four machine-learning models).*

Set	Policy	Window	Liquidated (% demand)	Inventory change (%)	Fill-rate change (pp)	Stockout change (%)	s* range
M5	Quantile	main	0.02	−0.1	−0.00	+0	Unstable (few units)
M5	Quantile	second	0.01	−0.1	−0.00	+0	Unstable (few units)
M5	Quantile	third	0.02	−0.2	−0.00	+0	Unstable (few units)
M5	Fixed	main	0.79	−2.3	−0.29	+7	1.06–1.32
M5	Fixed	second	1.08	−3.0	−0.20	+5	1.01–1.18
M5	Fixed	third	1.21	−2.7	−0.29	+7	1.03–1.23
M5	Dead13	main	0.67	−2.5	−0.65	+13	1.35–2.08
M5	Dead13	second	0.82	−4.9	−0.46	+10	1.21–1.66
M5	Dead13	third	0.90	−3.3	−0.54	+13	1.19–1.70
VN1	Quantile	main	1.18	−4.3	−0.02	+0	0.91–1.04
VN1	Quantile	second	0.27	−1.7	−0.02	+1	0.92–1.21
VN1	Quantile	third	1.74	−7.4	−0.02	+0	0.91–1.04
VN1	Fixed	main	2.76	−6.1	−0.67	+20	1.00–1.22
VN1	Fixed	second	1.47	−5.0	−0.11	+18	0.97–1.07
VN1	Fixed	third	1.91	−4.4	−0.62	+17	1.03–1.29
VN1	Dead13	main	1.21	−4.0	−0.09	+23	1.03–1.37
VN1	Dead13	second	0.76	−3.8	−0.08	+22	1.06–1.32
VN1	Dead13	third	0.47	−2.3	−0.07	+18	1.16–1.60

The quantile liquidation rule reduces realised fill rate by at most 0.02 percentage points with LGB-Q in every dataset–window combination. It almost never activates on M5 and reduces inventory by 1.7–7.4% on VN1. The fixed and dead13 policies incur larger service losses, increasing the stockout rate by 5–20% and 10–23%, respectively (Table 6).

Under the assumed cost specification, none of the liquidation policies demonstrates a consistent economic benefit over the 26-week horizon. For the VN1 quantile rule, the break-even salvage ratios range from 0.91 to 1.21 times unit cost; the corresponding lower bound for the remaining evaluated rules is 0.97. These are model-based thresholds, not observed resale proceeds or realised profits.

5.7 Sensitivity to dataset differences

We examine three potential sources of dataset-specific results (further details are reported in the repository). Reweighting VN1 series to match the M5 demand-feature distribution reduces the mean standardised difference from 0.32–0.39 to 0.02–0.03, without changing the reported VN1 conclusions. Removing price features from the VN1 global models, for which price is observed in fewer than half of the weeks, and restricting M5 models to the 21 features shared with VN1 each change the inventory required by LGB-Q by less than 1% across the tested windows. These checks do not explain the observed differences through the examined feature-distribution and preprocessing variations. Other differences between the datasets remain possible.

5.8 Robustness checks and a neural baseline

Hyper-parameters. We refitted LGB-Q and LGB-T (h = 3, main window) under three alternative LightGBM settings: small trees (15 leaves, at least 100 samples per leaf), large trees (255 leaves, at least 50) and a regularised setting (31 leaves, at least 500, L2 penalty 10). On M5, LGB-Q requires 4.7–6.8% less inventory than LGB-T and 7.5–11.0% less than TSB-NB at a fill rate of 0.92 under all four settings, with every interval above zero, and its own inventory varies by less than 3% across settings. On VN1, the conclusions of Section 5.2 hold under three of the four settings. With small trees, LGB-Q no longer requires significantly less inventory than TSB-NB, LGB-T needs 8.9% less inventory than LGB-Q at a fill rate of 0.92, and LGB-Q itself needs 6.3% more inventory at 0.94 than with the default setting.

Lead time. We repeated the equal-fill comparison with lead times of one and four weeks (forecast horizons 2 and 5) in all three windows, using the same methods except HGB-Q. On M5, in all nine combinations of window and lead time, LGB-Q requires significantly less inventory than LGB-T (2.5–7.4%), LGB-C (4.5–10.9%), TSB-NB (4.9–14.8%) and ETS

(5.9–13.4%) at a fill rate of 0.94, and its regret never exceeds 0.2%. On VN1, C2 requires 3.2–10.9% less inventory than LGB-Q at a fill rate of 0.92 in all nine combinations, with the interval excluding zero in seven. Over the 18 runs, the maximum regret relative to the best of these six single methods is 2.0% for C2 and 12.6% for LGB-Q, the latter in the second VN1 window with a four-week lead time.

Neural baseline. Outside the seven-method benchmark, we ran a univariate N-HiTS network [26] with a multi-quantile loss under the same protocol: direct quantiles of h-week totals, refits every 13 weeks and fixed hyper-parameters. Its wQL lies between those of LGB-Q and TSB-NB in every run (M5: 0.078, 0.083, 0.089; VN1: 0.107, 0.079, 0.104), and its upper quantiles are well calibrated (empirical coverage of the 0.99 quantile 0.986–0.992). It is not the most inventory-efficient method in any run: on M5 it requires 2.6–10.0% more inventory than LGB-Q at fill rates 0.90–0.94, and on VN1 its frontier rank among eight single methods is 2.8–4.4. With N-HiTS included, C2 keeps the lowest maximum regret (0.9%). We did not evaluate further neural architectures such as DeepAR or PatchTST; with a CPU-only budget, N-HiTS serves as one representative global neural forecaster, not as an upper bound for deep learning.

6 DISCUSSION

Forecast methods should be compared at the same realised service outcome, not merely at a common nominal quantile level. On M5, setting $\tau = 0.90$ produces fill rates from 0.935 to 0.967 across methods, reflecting calibration differences. Rankings at equal τ therefore need not measure the inventory required for equivalent service. The variation in VN1 method rankings across the three windows likewise indicates that results from a single evaluation period may not transfer to adjacent periods, consistent with the concern raised by Theodorou et al. [7].

On VN1, inventory performance is associated with realised demand changes. TSB-NB requires less inventory for the stable-demand subset, while LGB-Q achieves a fill rate of 0.92 on growing-demand series in two windows where TSB-NB does not. Theodorou et al. [7] also report divergence between accuracy and inventory costs, particularly when lost sales are heavily penalised. Our comparisons show that the inventory ranking can additionally change across periods and that this behaviour is not observed on M5 at the fill rates examined here. A hypothetical rule with advance knowledge of which series will grow would require 5.6–9.8% less inventory than LGB-Q on VN1. A classifier trained on pre-test features retrieves part of this advantage but produces unstable results (see repository); the ex-post classification itself cannot support a deployable decision rule.

Direct quantile estimation is relevant to the comparison. At a fill rate of 0.94 on M5, the point-forecast-plus-safety-stock and conformal approaches (LGB-T and LGB-C) require 3.3–8.2% more inventory than directly estimated LightGBM quantiles. Differences in forecast construction and aggregation level may contribute to contrasts with [7], which used empirical residual quantiles, and with the daily simulation in [23], where empirical methods performed well. These studies do not isolate those factors, so the comparison should not be treated as a causal explanation.

Across the six dataset–window runs, C2 has the lowest reported maximum regret under the specified metric. Regret is defined for each run as excess inventory over the best single method, averaged across the fill-rate targets a method reaches. The maximum is 0.9% for C2, compared with 9.4% for LGB-Q, 9.5% for LGB-T (which does not reach the highest fill-rate targets in any run), 11.4% for HGB-Q, and 20.7% for TSB-NB. When a lost sale is valued at no more than 200 weeks of holding cost, C2 stays within 3.1% of the cheapest single model in each run. With higher lost-sales penalties, all methods operate at $\tau = 0.99$; LGB-Q then attains the highest fill rate, and C2 can cost up to 29% more. Because C2 was selected after examining the same experiments, these descriptive regret figures should not be interpreted as prospective performance guarantees.

Limitations. Recorded sales proxy demand, and historical stockouts cannot be identified. Hyperparameters are not tuned; the M5 comparisons persist under four LightGBM settings and the VN1 comparisons under three of four (Section 5.8). Lead times of one, two and four weeks were examined in every window, but without HGB-Q. Demand-growth groups are defined ex post. The bootstrap intervals resample series rather than periods, and only three windows are evaluated per dataset. Differences between datasets also limit generalisation to other retailers. A supplementary repository experiment reports that zero-shot Chronos-2 [22] has lower accuracy than LGB-Q on 69–73% of VN1 series and is not the most inventory-efficient method; this result and the N-HiTS baseline (Section 5.8) are not part of the seven-method main benchmark.

7 CONCLUSION

We compared seven probabilistic forecasting methods under one replenishment and liquidation policy on M5 and VN1 over three 26-week test windows. On M5, LightGBM quantile regression (LGB-Q) has the lowest volume-weighted quantile loss and requires 3–11% less inventory than point forecasts with safety stock, conformal quantiles, TSB-NB, and exponential smoothing at a realised fill rate of 0.94 in all three windows.

On VN1, method rankings differ across windows. TSB-NB uses less inventory for stable-demand series, whereas LGB-Q reaches the evaluated fill-rate target for growing-demand series in two windows where TSB-NB does not. Neither ADI–CV² grouping nor the tested unsupervised segmentation consistently improves prospective method selection. A post hoc equal-weight average of LGB-Q and TSB-NB quantiles requires 6–11% less inventory than LGB-Q on VN1 at a fill rate of 0.92 and remains within 1.5% on M5. This result needs independent validation because the model pair was selected after the component results were examined.

Quantile-based liquidation reduces inventory on VN1 with changes in fill rate of no more than 0.02 percentage points; however, no policy establishes a consistent economic advantage over the 26-week horizon under the assumed costs. Subsequent evaluations should validate quantile combinations on new datasets and periods, assess model selection using pre-test indicators of demand growth, extend the liquidation horizon, and test additional forecasting models under the same decision policy.

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